

MAGTUNE MAPPER PRO | User Guide

The Professional Mapping for Luthiers

1. INTRODUCTION

The **Magtune Mapper Pro** is the dedicated software interface for your Magtune Precision Magnetic Gauge. It allows you to transform raw measurement data into high-resolution topographic maps, organic instrument outlines, and professional workshop reports.

Deployment Options:

- **Desktop App:** Run the standalone MagtunePro.exe for a native Windows experience.
 - **Web Version:** Open the Magtune_Mapper.html file in any modern browser (Chrome or Edge recommended).
 - **Offline Mode:** Both versions are 100% functional without an internet connection.
-

2. INTERFACE OVERVIEW

The Sidebar (Left)

- **Import Image:** Load a clear, top-down photo of the instrument.
- **Metadata Fields:** Enter the Guitar Model, Serial Number, and Technician Notes.
- **Toolbox:** Access Heatmap controls, Trace mode, and Smoothing.
- **Export Center:** Generate PDF reports or CAD-ready DXF/SVG files.

The Workspace (Center)

This is your interactive mapping area. It supports high-resolution images with a precision coordinate system.

Logged Points (Right)

A real-time list of every measurement taken. Each entry is numbered and color-coded to match the visual markers on the instrument.

3. NAVIGATION (ZOOM & PAN)

To ensure pinpoint accuracy when tracing or marking, use the navigation engine:

- **Zoom In/Out:** Hover your cursor over an area and **Scroll the Mouse Wheel**. The view will zoom toward your cursor.
 - **Panning (Moving):** Hold the **Middle Mouse Button** (or **Alt + Left Click**) and drag the image.
 - **Reset View:** Click the **[FIT]** button in the bottom-right cluster to snap the image back to its original size.
 - **Rotate:** Click the **[U]** button to rotate the image in 90° increments to match your workspace orientation.
-

4. THE WORKFLOW: STEP-BY-STEP

Step 1: Instrument Tracing

Before measuring, define the instrument's boundary so the heatmap stays within the wood.

1. Click **[TRACE: OFF]** to turn it **ON** (Green).
2. Click points around the perimeter of the guitar. A purple dotted line will follow your path.
3. You can **click and drag** any purple point to fine-tune the outline.
4. Once finished, click **[TRACE: ON]** to close the path and turn it **OFF**.

Step 2: Smoothing (Optional)

If your trace is jagged, click [**SMOOTH CURVE: OFF**]. The app will apply a Cardinal Spline algorithm to transform the trace into an organic, flowing shape.

Step 3: Logging Measurements

1. Ensure the Magtune Pro App is the active window on your PC.
2. Click a spot on the guitar image. A **yellow "?" marker** will appear.
3. Press the **HOLD** button on your physical Magtune device.
4. The value (e.g., 2.34) is "typed" into the app. The marker turns blue and is numbered (e.g., ①).
5. The data appears instantly in the **Logged Points** sidebar on the right.

Step 4: Generating the Heatmap

1. Once you have several points, click [**HEATMAP: OFF**] to turn it **ON**.
 2. The app calculates a smooth color gradient across the wood surface.
 -  **Blue:** Thinnest areas detected.
 -  **Green/Yellow:** Mid-range/Average thickness.
 -  **Orange/Red:** Thickest areas.
 3. Adjust the **HEATMAP OPACITY** slider to see the wood grain or internal bracing through the colors.
-

5. DATA MANAGEMENT

- **Correcting Position:** You can click and drag any blue marker to reposition it on the image.
- **Deleting Points:** Click the red [**X**] next to any measurement in the right sidebar to remove it.

- **Undo/Redo:** Use the Blue **[UNDO]** and **[REDO]** buttons to revert accidental changes to your markers or trace lines.
-

6. EXPORTING REPORTS & CAD

Professional PDF Report

Click **[EXPORT PDF REPORT]** to generate a formal document.

- Includes the instrument map with all markers and the Heatmap.
- The Sidebar contains the Date (**DD/MM/YYYY**), Model, and Serial Number.
- Measurements are listed in an organized **3-column table**, allowing up to 90+ points on a single page.

CAD Export (DXF & SVG)

For luthiers using CNC or Laser cutters, use the **[DXF]** or **[SVG]** buttons.

- **What is exported:** Only the **Instrument Trace** (Outline).
 - Measurement markers are automatically excluded to provide a clean vector path for template making.
-